Sam Park

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Projects

Drumbeat Atlas, Drumbeat Labs (Capstone) **2026**

- Architected and deployed a social media analysis pipeline on Cloudflare and GitHub Actions to collect a 37,000-post corpus across several campaign topics
- Transcribed spoken audio and on-screen text of each post for text embeddings, clustering, and sentiment scoring
- Found views are driven by follower count and post age while engagement is associated with semantic content
- Trained embedding probes using logistic regression for five content categories at held-out AUC 0.80 to 0.91, finding posts that name wrongdoing draw the strongest engagement in two sponsored campaigns
- Identified a recency bias in the Scrape Creators API and scoped temporal claims accordingly

Drumbeat Viewer, Drumbeat Labs (Capstone) **2026**

- Built a web interface in React for post embeddings, sentiment, and metadata to support real-time exploratory demos
- Renders the UMAP embedding projection client-side via DuckDB-wasm + Mosaic + vgplot so that all computations for selections and crossfilters occur instantly in-browser
- Links density distribution plots that crossfilter the map, with interval brushes that highlight the selection
- Reports live deviation from corpus mean and feature correlations over the current selection in a heatmap strip

BEDbase + gtars, University of Virginia **2026**

- Built an updated web interface in React for BEDbase, incorporating an interactive UMAP embedding viewer
- Live-rendered Observable Plot visualizations replace our precomputed ones from ggplot2
- Drove the porting of the lab's algorithms into Rust + WASM bindings to facilitate these features

Genomic Regions, University of Virginia **2026**

- Explored spatial relationships across one million+ pretrained Region2Vec genomic region embeddings
- Annotated genomic regions by transcription factor binding across three cell lines, recovering erythroid, hepatocyte, and B-cell programs unsupervised
- Built an interactive multi-panel explorer in React for region embeddings incorporating linked UMAPs, chromosome distributions, and region co-occurrence plots

Experience

Bioinformatician, University of Virginia **2024 – Current**

- Built interactive user and data visualization interfaces for tools published by Sheffield Lab
- Developed Refget SCOM interface to visualize comparisons between genomic sequence collections via Vega-Lite
- Ran stratified GWAS and pQTL colocalization analyses to identify credible signals for sex differences in lung function
- Investigated markers of abnormal lobe function and transplant rejection using single-cell gene expression in human lung
- Engineered a single-cell RNAseq pipeline for agentic AI integration with canonical configs, machine-readable outputs, and pipe-friendly I/O
- Contributed to pilot study using biological age prediction and GWAS to explore genetic sources of biological age gap within human proteome and transcriptome

Automation Engineer, Merck & Co. (Contract) **2022 – 2025**

- Helped lead automation community project incorporating R Shiny, plumber API, AWS, and PI Web API to automate manual continuous historian report writing process, eliminating 500+ manually-written reports per year
- Developed initial proof of concept for continuous historian report tool that served as foundation for production codebase
- Worked on automated data pipelines that use R Markdown and PI Web API to manage factory data on AWS S3 and RDS

Skills

Languages & Data: R, Python, JavaScript, TypeScript, SQL
Statistics & Measurement: permutation testing, effect estimation, linear regression
Machine Learning: text embeddings, supervised classification and probing, PCA, UMAP, clustering, random forest
Data Visualization: Vega-Lite, Mosaic/vgplot, Observable Plot, D3, ggplot2, Tableau
Infrastructure: React, FastAPI, Cloudflare, GitHub Actions, DuckDB, AWS, R Shiny, plumber API, Streamlit

Education

University of Virginia
Master of Science (Aug 2026). Program: Data Science
Bachelor of Science (May 2022). Major: Biomedical Engineering | Minor: Computer Science